SHA-1 RTL optimization on VTR

Auditable primary PPA estimate · open 45 nm FPGA proxy

Three cycle-equivalent RTL transformations improve the paired area-delay-power estimate by 2.27%

Prepared by Göther Labs

Measurement completed: 9 August 2026

Academic open-FPGA estimate. Not ASIC signoff, a commercial FPGA result, or manufactured silicon.

1. Executive result

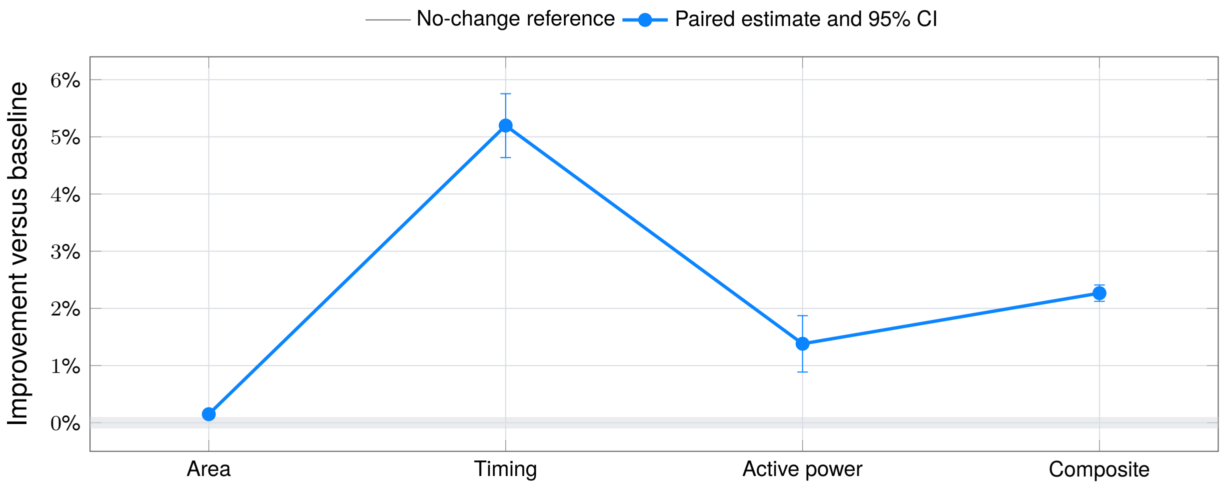
CORRECTED BASELINE VERSUS ACCEPTED RTL

The accepted RTL improves the equal-weight area-delay-active-power estimator by 2.27% (95% CI: 2.12% to 2.41%) without changing the interface, defined-input cycle behaviour, latency or synthesized register count.

Primary metric	Baseline median	Accepted median	Paired improvement	95% CI / state
Total area	16,614,693 MWTA	16,614,693 MWTA	**0.15%**	0.08% to 0.21% · **better**
Critical path	15.0054 ns	14.2090 ns	**5.20%**	4.64% to 5.75% · **better**
Active total power	9.9125 mW	9.7755 mW	**1.38%**	0.89% to 1.87% · **better**
Composite estimator	1.000000	0.977335	**2.27%**	2.12% to 2.41% · **better**

Paired primary PPA improvement

64 fixed certification seeds · lower is better · bars are two-sided 95% confidence intervals



The reported composite value 0.977335 is the paired geometric-mean estimator, not a median. The median of the 64 per-seed composite ratios is 0.978469. Both summarize the same fixed sample but answer different questions; the estimator and its paired interval define acceptance.

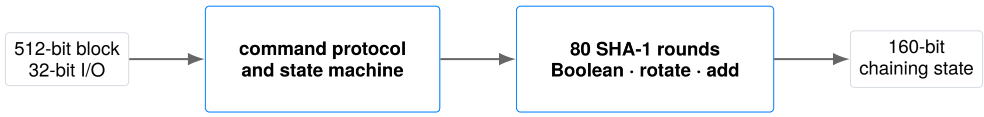
2. Module and benchmark

WHAT IS BEING OPTIMIZED

The evaluated module is a sequential SHA-1 compression implementation from the public Verilog-to-Routing repository. The exact upstream source is the pinned `sha.v` at commit `95f5c6de…`.

It receives a 512-bit block through a 32-bit command/data interface and updates a 160-bit chaining state over 80 SHA-1 rounds. Only `sha.v` is editable; reset, command protocol, read order, latency, throughput and observable defined-input cycle behaviour remain fixed.

```
module sha1(clk_i, rst_i, text_i, text_o, cmd_i, cmd_w_i, cmd_o);
  input clk_i, rst_i, cmd_w_i;
  input  [31:0] text_i;
  output [31:0] text_o;
  input  [3:0]  cmd_i;
  output [3:0]  cmd_o;
endmodule
```



The original VTR artifact does not produce the standard digest for `abc`. A separate historical conformance correction created the frozen reference:

$$\text{SHA1}(\texttt{abc}) = \texttt{a9993e364706816aba3e25717850c26c9cd0d89d}.$$

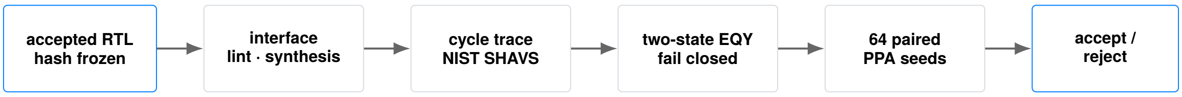
The correction is excluded from the optimization claim. Every ratio compares the accepted RTL with this corrected baseline.

Artifact	SHA-256
Corrected baseline `sha.v`	`191a4f2148a4efda7aadd24480eb13d78a1d2c0c7e8a3fcc37c44f6a8e8011e5`
Accepted `sha.v`	`ea63b97faf69b14409a3077cd3357205c0e044b277d64537b98a4a4773a4f2e7`

3. Evaluation contract

FROZEN BEFORE THE ACCEPTED MEASUREMENT

Dimension	Frozen choice
Editable artifact	`sha.v` only; submissions frozen by SHA-256.
Functional behaviour	Exact interface, reset, protocol, latency, output order and defined-input cycle behaviour.
Formal semantics	Two-state equivalence after the declared reset; inputs are defined, not X/Z.
Conformance	NIST SHA-1 Short and Long Message corpus; 129 cases.
Formal tool	EQY commit `6734d8c2`…; fail closed; 2 CPU, 7 GB, 10 min.
Physical flow	VTR/VPR commit `95f5c6de`…; pinned Linux/amd64 image.
Architecture	`k6_N10_I40_Fi6_L4_frac0_ff1_45nm.xml`.
Power	VTR PTM45, 0.9 V, 85 °C; fixed active and idle traces.
Search sample	Seeds 1, 7, 19, 43, 97; provisional ranking only.
Certification sample	Remaining 64 seeds in 1–68; disjoint, fixed $n = 64$, no extension.
Primary score	Equal-weight paired area, delay and active-total-power geometric mean.
Secondary metric	Energy per completed block; excluded from the primary score.



Exact decision rule. All correctness and evidence-integrity gates must pass. The composite one-sided 95% upper confidence bound must be below 1.0, and no primary metric may have a one-sided 95% lower bound above 1.0. Otherwise the result is inconclusive or regressed. No undefined “materiality” threshold is used.

4. Verification before PPA

REPLAYABLE EVIDENCE FOR THE ACCEPTED HASH

Gate	Public replay evidence	Result
Structural policy	Interface record, lint logs and synthesis outputs	PASS
Cycle regression	Compiled cycle harness and complete run logs	PASS
NIST SHAVS	Frozen 129-case corpus, harness and run log	PASS
Sequential EQY	Configuration, strategies, solver logs and PASS marker	PASS
Physical integrity	Every VTR/ACE output for both designs and all 64 seeds	PASS

The formal proof is scoped to two-state, defined-input hardware semantics after the declared reset. It compares the complete sequential design and all selected cut points. A timeout or inconclusive result is a failure. This is deliberately narrower than claiming identical Verilog X/Z propagation; that distinction matters for the majority rewrite and is stated rather than hidden.

The public release asset contains both raw evidence trees, the exact accepted and baseline records, the formal and functional inputs, tool-flow sources and an internal manifest for every archived member. The compact repository checks arithmetic and identity. The full-evidence mode additionally verifies all 11,362 archived files, validates declared public transformations, derives every certification metric from all 128 raw VTR/ACE run trees, compares them with both archived records and compact rows, and binds every measured run to the corresponding RTL hash.

The bundle records one documentation-only correction with source and public hashes: stale `cmd_i[2:0]` prose in the source flow manifest is published as the frozen `cmd_i[3:0]` RTL interface. The measurement RTL and every raw output remain unchanged.

```
python3 verify.py
python3 verify.py --evidence-archive \
  primary-ppa-full-evidence.tar.gz
```

> **Evidence boundary.** “Compact package consistency: PASS” does not certify raw EDA provenance. Only the second command may report “Full raw evidence: PASS”.

The 500 deterministic MCY mutations qualify the verification stack: 454 are detected by simulation, 28 additional distinguishable mutations are rejected only by formal, and 18 are proved equivalent. These qualification results are included with the baseline evidence and are not counted as candidate improvements.

5. Exact baseline-to-accepted diff

ALL CHANGED LINES FROM SHA.V

The patch below is generated directly from the two checked-in RTL files. No testbench, constraint, architecture, activity, parser or tool flag is editable by the candidate.

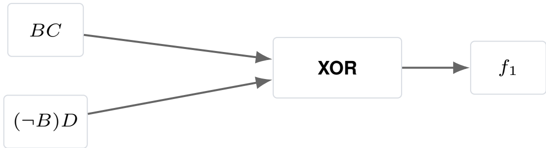
```
--- rtl/baseline/sha.v
+++ rtl/accepted/sha.v
@@ -121,20 +121,23 @@
        end

        // Hash functions
-       wire [31:0] SHA1_f1_BCD,SHA1_f2_BCD,SHA1_f3_BCD,SHA1_Wt_1;
-       wire [31:0] SHA1_ft_BCD;
+       wire [31:0] SHA1_B_xor_D,SHA1_C_xor_D;
+       wire [31:0] SHA1_f1_BCD,SHA1_core_f2,SHA1_core_f3,SHA1_core_ft;
+       wire [31:0] SHA1_Wt_1,SHA1_ft_BCD;
        wire [31:0] next_Wt,next_A,next_C;
-       wire [159:0] SHA1_result;

-       assign SHA1_f1_BCD = (B & C) ^ (~B & D);
-       assign SHA1_f2_BCD = B ^ C ^ D;
-       assign SHA1_f3_BCD = (B & C) ^ (C & D) ^ (B & D);
+       assign SHA1_B_xor_D = B ^ D;
+       assign SHA1_C_xor_D = C ^ D;
+       assign SHA1_f1_BCD = (B & C) | (~B & D);
+       assign SHA1_core_f2 = SHA1_B_xor_D ^ SHA1_C_xor_D;
+       assign SHA1_core_f3 = SHA1_B_xor_D & SHA1_C_xor_D;

        // The state machine value is one-based while executing SHA-1 rounds:
        // round=1 computes algorithm round 0 and round=80 computes round 79.
+       assign SHA1_core_ft = (round <= 7'd40) ? SHA1_core_f2 :
+                               (round <= 7'd60) ? SHA1_core_f3 : SHA1_core_f2;
        assign SHA1_ft_BCD = (round <= 7'd20) ? SHA1_f1_BCD :
-                            (round <= 7'd40) ? SHA1_f2_BCD :
-                            (round <= 7'd60) ? SHA1_f3_BCD : SHA1_f2_BCD;
+                              (D ^ SHA1_core_ft);

        // Odin II doesn't support binary operations inside concatenations presently.
        //assign SHA1_Wt_1 = {W13 ^ W8 ^ W2 ^ W0};
@@ -143,8 +146,6 @@
        assign next_Wt = {SHA1_Wt_1[30:0],SHA1_Wt_1[31]};       // NSA fix added
        assign next_A = {A[26:0],A[31:27]} + SHA1_ft_BCD + E + Kt + Wt;
        assign next_C = {B[1:0],B[31:2]};
-
-       assign SHA1_result    = {A,B,C,D,E};

        assign round_plus_1 = round + 1;

@@ -2156,11 +2157,11 @@
                        if (~busy)
                        begin
                                case (read_counter)
-                                       3'b100: text_o <= SHA1_result[5*32-1:4*32];
-                                       3'b011: text_o <= SHA1_result[4*32-1:3*32];
-                                       3'b010: text_o <= SHA1_result[3*32-1:2*32];
-                                       3'b001: text_o <= SHA1_result[2*32-1:1*32];
-                                       3'b000: text_o <= SHA1_result[1*32-1:0*32];
+                                       3'b100: text_o <= A;
+                                       3'b011: text_o <= B;
+                                       3'b010: text_o <= C;
+                                       3'b001: text_o <= D;
+                                       3'b000: text_o <= E;
                                        default:text_o <= 32'b0;
                                endcase
                                if (|read_counter)
```

Attribution boundary. The measured PPA result belongs to the complete patch. The following pages explain the three conceptual transformations, but no exact fraction of the gain is assigned to an individual statement.

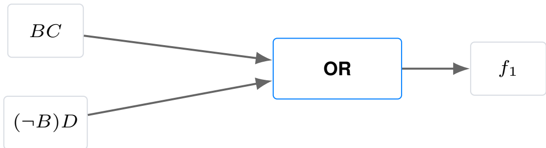
6. Rewrite 1: SHA-1 choose function

SOURCE: SHA.V:130

Corrected baseline



Accepted RTL



Why the expressions are equivalent

$$\begin{aligned} F &= BC \oplus (\neg B)D \\ &= BC \vee (\neg B)D. \end{aligned}$$

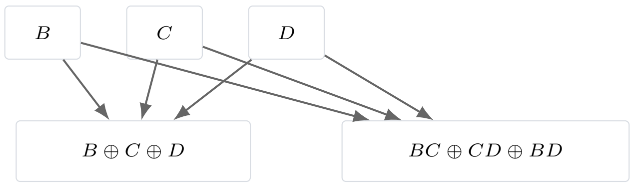
The two products are mutually exclusive. If $BC = 1$, then $B = 1$ and $(\neg B)D = 0$; if $(\neg B)D = 1$, then $B = 0$ and $BC = 0$. XOR and OR therefore agree for every binary input. This preserves the SHA-1 choose function: select C when $B = 1$, otherwise select D.

> **Formal result.** EQY proves the replacement in the complete two-state sequential circuit under the declared reset and defined-input model.

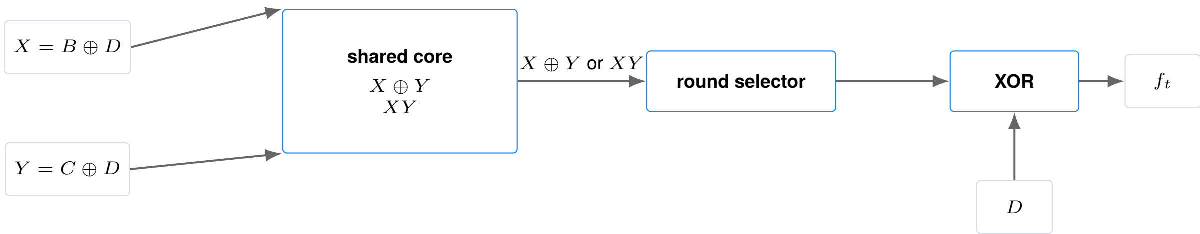
7. Rewrite 2: shared SHA-1 Boolean core

SOURCE: SHA.V:126–141

Corrected baseline · separate parity and majority cones



Accepted RTL · shared XOR terms and round-selected core



Let $X = B \oplus D$ and $Y = C \oplus D$. For parity rounds,

$$D \oplus (X \oplus Y) = D \oplus B \oplus D \oplus C \oplus D = B \oplus C \oplus D.$$

For majority rounds, use the value of D:

$$D \oplus (XY) = \begin{cases} BC, & D = 0, \\ B \vee C, & D = 1. \end{cases}$$

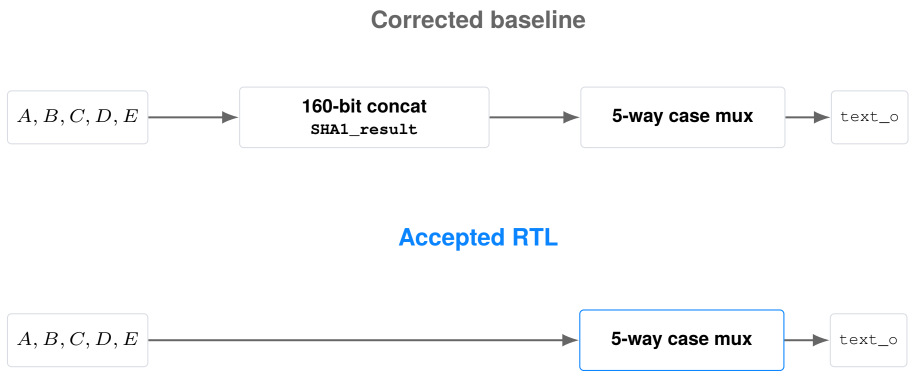
Those are exactly the two cofactors of Maj(B, C, D). The new form shares $B \oplus D$ and $C \oplus D$ across the parity and majority paths and moves the final D XOR after the round selector.

> **Four-state caveat.** The identity is exact for binary hardware values. It intentionally does not claim identical X/Z propagation in an event-driven Verilog simulation. The formal and functional contract supplies defined binary inputs after reset.

8. Rewrite 3: direct chaining-state readout

SOURCE: SHA.V:127 AND SHA.V:2160–2164

The baseline first creates a 160-bit concatenation and then uses a five-way `case` mux to read constant 32-bit slices. The accepted RTL retains the same case conditions and assignments but connects the five state registers directly.



For each `read_counter` value, concatenation slicing is an alias:

$$\{A,B,C,D,E\}[159{:}128] = A, \quad \{A,B,C,D,E\}[127{:}96] = B, \quad \ldots \quad \{A,B,C,D,E\}[31{:}0] = E.$$

The output mux, register values, read order and cycle timing are unchanged. This diagram shows the actual five-way case topology; it does not depict a variable part-select.

> **Formal result.** EQY proves that every output word is identical on every compared cycle under the declared model.

9. PPA environment and statistics

OPEN 45 NM FPGA PROXY

The target is VTR's homogeneous `k6_N10_I40_Fi6_L4_frac0_ff1_45nm` architecture: six-input LUTs clustered ten per logic block, with the architecture's routing and timing model. Power uses PTM45 properties at 0.9 V and 85 °C. MWTA is minimum-width-transistor area, not square-micrometre physical area.

Two 5,000-cycle ACE traces are frozen and hashed. The active trace continuously executes legal blocks and completes 60 blocks; the idle trace keeps the clock active after reset with an inactive interface. Active and idle power analyses reuse the same placement and route for a given design and seed.

The primary per-seed score is

$$r_s = \sqrt[3]{r_{A,s}\, r_{D,s}\, r_{P,s}}, \qquad \widehat{r} = \exp\left(\frac{1}{64}\sum_{s\in S}\log r_s\right),$$

where A is total MWTA, D is post-route critical-path delay and P is active total power. Energy is secondary:

$$E_{\text{block}} = \frac{P_{\text{active,total}}\, D\,(5{,}000 \text{ cycles})}{60 \text{ completed blocks}}.$$

Two-sided 95% Student-t intervals use 63 degrees of freedom on paired log ratios. The one-sided bounds in the decision rule use the same predeclared fixed sample. These intervals quantify variation across the selected implementation seeds; they do not cover process, voltage, temperature, workload or tool-version uncertainty.

Seed policy. Search uses only 1, 7, 19, 43 and 97. Certification uses the 64 values in 1–68 that are disjoint from that search set. The pool, $n = 64$ stopping rule and confidence level are part of the domain contract and are not selected after seeing this candidate.

10. Every paired certification seed

PRIMARY PPA DISTRIBUTIONS

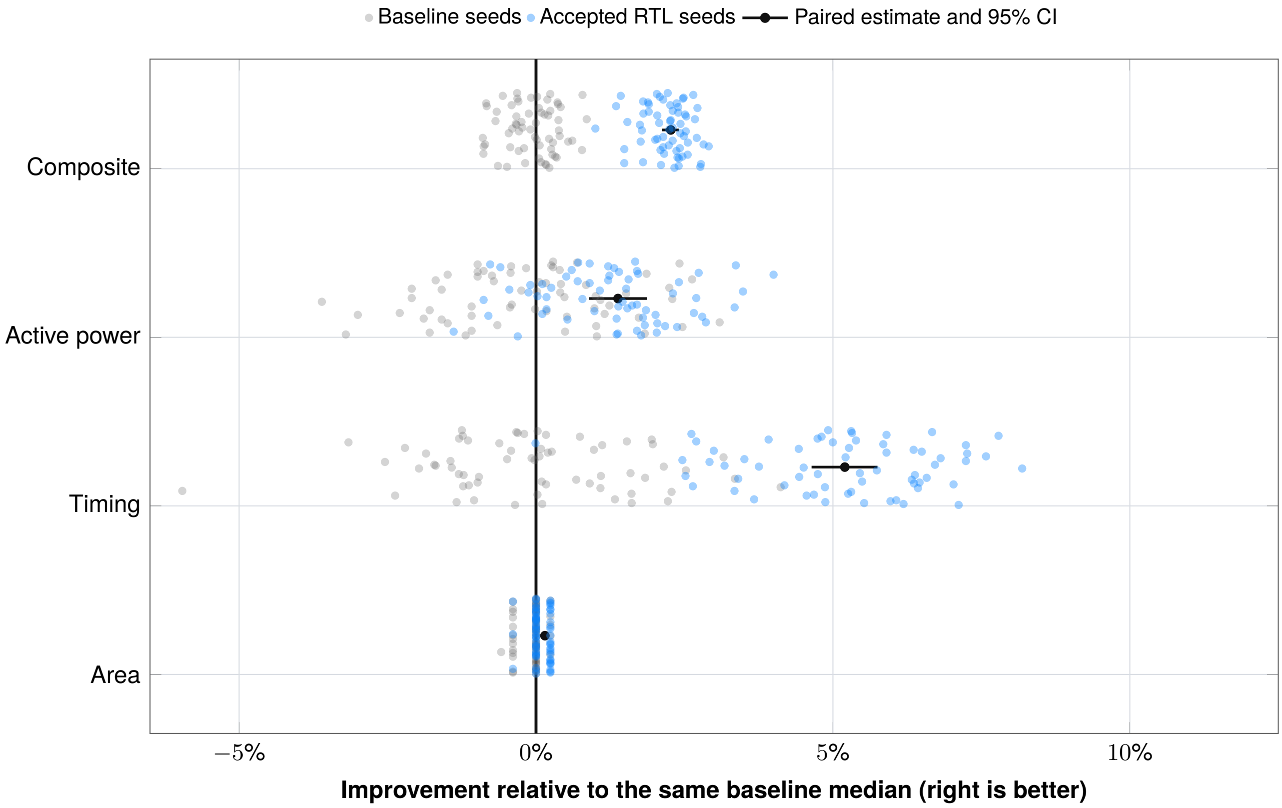


FIGURE 2 · OVERLAID BASELINE AND ACCEPTED SEED CLOUDS

Gray and blue points are absolute results normalized to the same baseline median. The same seed receives the same vertical jitter. Black points and segments show the paired estimator and interval that determine confidence.

Metric	Wins	Ties	Losses
Area	30	28	6
Timing	64	0	0
Active power	46	0	18
Composite	64	0	0

11. Secondary metrics and evidence identity

MEASURED CONSEQUENCES AND PUBLIC AUDIT TRAIL

Metric	Baseline median	Accepted median	Interpretation
Energy / block	12.3896 nJ	11.5718 nJ	6.51% paired improvement (95% CI 6.14% to 6.87%)
Fmax	66.6426 MHz	70.3781 MHz	Reciprocal of routed critical path
Packed CLBs	188	182	Six fewer logic blocks in every synthesized candidate netlist
Registers	892	892	Unchanged
Idle total power	4.4670 mW	4.3490 mW	Informative; excluded from score

The energy result is larger than the active-power result because the accepted implementation is both faster and slightly lower-power. CLB, register, Fmax and idle-power fields are published per seed and recomputed by the verifier; they are not unsupported summary annotations.

Public evidence object	SHA-256 / identity
Accepted certification record	`12a3b40974d760898d583cf85768602f8e1d5a0ba3d771d60ff88940002f30dd`
EQY PASS marker	`9aeb6fa6171b28a9ae33345754b973b407317736d6ac333dd1e4a45d07c575be`
EQY driver log	`2836ffc302152cb420ca6b2855205460ce6e6b5b1d8312c1ec6fa02e9f4f810a`
Full evidence release asset	`fd345d155d5f2225d3a3a6e9f0431b9db4b0f0dcc1d1951083c53a7997821704`
Archive size	145,901,014 bytes

The release asset is available from the public `v1.3.0-rc4` release. Its internal manifest covers every member; the Git release and this report provide the external content anchor.

12. Reproduction and verifier trust model

WHAT EACH COMMAND ESTABLISHES

The repository intentionally separates three levels of evidence:

1. **Git identity**: checksums cover the exact compact package and the commit records its history.
2. **Compact consistency**: `python3 verify.py` validates strict schemas, RTL hashes, the exact patch, all 64 rows, auxiliary fields, formulas, confidence intervals and acceptance rules.
3. **Raw-evidence audit**: supplying the release archive verifies its external digest, internal manifest, all 11,362 files, declared transformations, formal/NIST markers and RTL identities, then independently parses every raw VTR/ACE certification run and binds every published metric to those logs.

```
curl -LO https://github.com/juan-fernandez-gotherlabs/\
rtl-optimization-case-study/releases/download/v1.3.0-rc4/\
primary-ppa-full-evidence.tar.gz

python3 verify.py --evidence-archive \
  primary-ppa-full-evidence.tar.gz
```

The verifier uses explicit checks rather than Python `assert`, rejects duplicate JSON keys, non-finite values, fractional seeds, unit-scale violations, incomplete manifests, altered auxiliary fields and rehashed raw-log substitutions, and behaves identically under `python -O`. Its adversarial unit suite exercises these cases in CI.

The included flow snapshot pins the Dockerfile, Python lock, VTR and EQY commits, architecture, PTM, activity vectors, harnesses and evaluator sources. A clean physical rerun still requires Docker with Linux/amd64 support, two CPUs, 7 GB RAM and the pinned source checkouts. Runtime networking is disabled.

Trust boundary. A checksum proves content identity, not authorship. The Git commit/tag and public release are the publication anchors. A customer deployment should add organisation-controlled signing or an attestation service to that release process.

13. Limits and customer transfer

WHAT THE RESULT DOES AND DOES NOT ESTABLISH

This report does not provide:

- ASIC synthesis, place-and-route or signoff;
- commercial-FPGA implementation data;
- extracted parasitics, PVT corners, clock-tree signoff, IR drop, EM, DRC/LVS, package, yield or production-test analysis;
- proof that the same edits improve another architecture, technology or workload;
- exact PPA attribution to one transformation;
- identical four-state X/Z propagation for the majority-core rewrite;
- a recommendation to use SHA-1 in a new security design.

The 64-seed interval models implementation-seed variability for the fixed VTR contract. It does not model silicon process variation or generalise automatically to a customer technology.

Transfer to a customer pilot. Replace the academic proxy with customer-owned RTL, formal assumptions, libraries, constraints, activity, tools and signoff policy. The transferable capability is a small inspectable source change evaluated by a contract that can justify acceptance to the engineers responsible for the design.

Primary references

- VTR/VPR documentation
- VTR power estimation
- EQY sequential equivalence checking
- NIST secure hashing validation
- Public case-study repository
- Göther Labs

Conclusion. Within the declared VTR 45 nm FPGA proxy and two-state contract, the accepted three-transformation patch passes the published functional and formal gates and improves the paired primary PPA estimator by 2.27% (95% CI 2.12% to 2.41%). The compact calculations and complete raw evidence are both publicly auditable.